Embedded-Text Leakage in Multimodal Meme Sentiment Classification: A Resource-Constrained Study of When Fusion Really Helps

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***Abstract*—**Prior controlled comparisons of text-only, image-only, and multimodal sentiment classification on social media report that fusion helps because text and images carry complementary signal, using photographic images where caption and picture are produced independently. Internet memes break that independence: meme text is rendered as pixels on the image, so an image-only model can, in principle, re-derive the text signal a text-only model already sees. We ask whether fusion gains reported for photographic images still hold for memes, and how much of any gain is attributable to this embedded-text channel. We run a controlled, same-split comparison on a stratified 3,500-sample subset (2,800/350/350) of the public Memotion dataset (SemEval-2020 Task 8; 6,992 memes total), collapsing 5-class sentiment to 3 classes. We evaluate TF-IDF+LogisticRegression and frozen-DistilBERT text models, a from-scratch CNN and frozen-ResNet18 image model, and a concatenation-fusion MLP — each trained twice: on the original ("sharp") memes, and on the same memes after a strong Gaussian blur that destroys fine text-scale pixel detail (objectively confirmed via a 99.9% drop in mean Laplacian-variance sharpness on a 150-image sample) while preserving coarse colour and composition. On sharp images, fusion reaches 52.6% test accuracy / 36.6% macro-F1, clearly ahead of the best unimodal model (TF-IDF text: 42.6% accuracy; DistilBERT text: 33.4% macro-F1). Blurring barely changes the image-only model (−1.4pp accuracy, −0.2pp macro-F1), but collapses most of the fusion gain (−10.9pp accuracy, −3.6pp macro-F1): blurred-image fusion performs almost identically to text-only (+2.3pp accuracy, −0.4pp macro-F1). This indicates the fusion benefit here is substantially mediated by the image channel's access to legible embedded text rather than independent visual sentiment cues — with direct implications for systems that assume image and text are independent evidence sources when images may themselves contain the text. All experiments ran on constrained consumer CPU hardware (no GPU, 8GB RAM).

***Index Terms*—**Multimodal sentiment analysis, Internet memes, social media analytics, feature fusion, DistilBERT, ResNet18, embedded text, modality leakage, resource-constrained machine learning.

I. INTRODUCTION

Multimodal sentiment analysis on social media typically assumes text and image are independently produced evidence sources whose combination can only help, or at worst add noise, precisely because they are independent. That assumption is reasonable for a tweet consisting of a caption plus an attached photograph. It is not obviously reasonable for an Internet meme, where the "image" frequently *is* the text: bold caption text is rendered directly onto the picture. A convolutional image encoder operating on raw pixels does not know, a priori, that a region of high-contrast, high-frequency pixels is text rather than visual texture — it may simply learn to exploit it, whether or not that was the intended visual signal. Consider a meme pairing a stock photograph with the bold caption "WHEN YOU FINALLY SUBMIT THE PROJECT REPORT" rendered directly on the image: a text-only model reads the caption and infers relief/positive sentiment from the words themselves; an "image-only" model, denied the caption field, may still recover much of the same signal simply by reading the same words off the pixels — it is not actually limited to the photograph's visual content the way an image-only model would be for an attached but textually separate photograph. This paper asks: (RQ1) does multimodal fusion still outperform unimodal baselines for meme sentiment classification, replicating the pattern reported for photographic images; and (RQ2) how much of any fusion gain depends specifically on the image channel reading embedded text, versus independent visual cues? We answer RQ2 with a direct ablation: destroy legible text-scale detail with an objectively validated Gaussian blur, re-extract image features, and re-train every image-dependent model, holding everything else fixed.

II. RELATED WORK

Controlled text-vs-image-vs-multimodal comparisons on real social-media data are rare. The closest located prior work performs a same-dataset, same-metrics comparison

of text-only, image-only, and multimodal sentiment classification on photographic Twitter image-text pairs, and reports that multimodal fusion outperforms either unimodal model on every aggregate metric, with the gain concentrated specifically on the subset of posts where the text-only and image-only models disagree, and a small regression on the subset where they already agree. That result is a natural baseline expectation for the present study, since it establishes what a "clean" complementary-modality fusion pattern looks like when text and image are produced independently. Multimodal fusion more broadly has a mature survey literature covering fusion-architecture taxonomies, benchmark datasets, and open challenges [1], [2], but these surveys largely treat "image" as a visually distinct channel from "text" without examining cases where the two are pixel-level coupled, as in memes.

The Memotion / SemEval-2020 Task 8 dataset used here was introduced explicitly to study sentiment, humor, sarcasm, offensiveness, and motivational content in memes as combined text-image artifacts, with the shared task's participating systems overwhelmingly adopting some form of text-plus-image fusion architecture [3]. The Hateful Memes Challenge similarly argues that meme meaning often depends on a genuine text-image interaction (e.g., a caption that is benign alone but harmful paired with a specific image) that neither modality captures in isolation, and was explicitly designed with "benign confounder" examples to force models away from relying on either modality alone [4]. Neither line of work, to our knowledge, directly quantifies how much of a meme multimodal model's measured cross-modal gain is attributable to the image channel simply re-reading the same embedded text a text-only model already sees, versus genuinely independent visual content (composition, colour, facial expression). This is precisely the gap this paper's blur ablation targets. Separately, the scene-text/TextVQA literature establishes that recognizing text rendered inside images is a major, distinct capability from general visual-object recognition, and that models without it systematically fail on image-understanding tasks where the answer depends on in-image text [5] — directly motivating the hypothesis that a meme image encoder's apparent "visual" sentiment signal may substantially be this same capability in disguise. Our text and image encoders follow established lightweight transfer-learning choices for resource-constrained settings: DistilBERT, a distilled, faster, smaller variant of BERT retaining most of its language-understanding performance [6], for text; and ResNet18, a compact residual convolutional architecture originally introduced for large-scale image classification and now a standard frozen feature extractor for transfer learning [7], for images — both used frozen (inference-only) throughout.

III. RESEARCH QUESTIONS

RQ1: Does feature-concatenation multimodal fusion outperform unimodal text-only and image-only sentiment classification on real Internet memes, as previously reported for photographic social-media images?

RQ2: How much of any fusion gain is attributable specifically to the image channel's access to legible embedded meme text, versus independent visual sentiment cues — tested by destroying legible text-scale detail via Gaussian blur and measuring the resulting change in each model's performance?

RQ3 (comparability check): Does the previously reported pattern that fusion's benefit concentrates on the text/image-disagreement subset replicate on meme data?

IV. DATASET

We use Memotion (SemEval-2020 Task 8 [3]): real, publicly shared Internet memes, human-annotated for 5-class sentiment, humor, sarcasm, offensiveness, and motivational content, with OCR-extracted, human-corrected caption text per meme. We accessed the data via the Ahren09/MMSoc_Memotion Hugging Face mirror (6,992 memes; 5,593/699/700 train/validation/test), disclosing this third-party-mirror provenance explicitly. We merge the 5-class sentiment label to 3 classes (very_positive+positive→positive; neutral→neutral; negative+very_negative→negative) for comparability with standard 3-class social-media sentiment framings.

Given this project's CPU-only, 8GB-RAM hardware (Section V), we draw a stratified, per-class-proportional subsample of 2,800/350/350 train/validation/test memes (3,500 total) rather than using the full 6,992, keeping the experiment tractable while preserving each split's class proportions. Resulting distribution — train: positive 1,659 (59.3%), neutral 882 (31.5%), negative 259 (9.3%); validation: positive 209 (59.7%), neutral 111 (31.7%), negative 30 (8.6%); test: positive 214 (61.1%), neutral 109 (31.1%), negative 27 (7.7%) — a real, substantial imbalance motivating macro-F1 as the primary metric.

V. METHODOLOGY

All models share the same split, fixed seed (42), and evaluation code. All image-dependent models are trained and evaluated in two parallel conditions, *sharp* and *blur*, from otherwise-identical pipelines.

A. Text and Image Pipelines

Text: (A1) TF-IDF (1–2 grams, 10k features) + class-weighted Logistic Regression on the corrected caption text; (A2) frozen DistilBERT mean-pooled embeddings (768-dim) + class-weighted Logistic Regression. Image: (B1) a small 4-layer CNN trained from scratch at 96×96 resolution; (B2) frozen ImageNet-pretrained ResNet18 penultimate-layer

embeddings (512-dim) + class-weighted Logistic Regression, run on both the sharp image and, as B2-Blur, the blurred image.

B. Blur Ablation

The blur condition applies a Gaussian blur (radius 14px, at native resolution before the standard 224×224 resize) to every image before feature extraction. We validate this destroys text-scale detail using an OCR-independent focus measure (variance of the image Laplacian) on a random 150-image test sample: mean sharp-image variance 2,021.6 vs. mean blurred-image variance 1.5 — a 99.9% reduction, i.e., fine high-frequency edge content that would carry legible character strokes is essentially eliminated, while coarse colour and compositional layout is preserved.

C. Multimodal Fusion

Feature-level concatenation of the A2 text embedding (768-dim) and the B2 image embedding (512-dim, sharp or blurred) → a 2-hidden-layer MLP (256→128→3) with dropout (0.4, 0.3), class-weighted cross-entropy, Adam (lr=1e-3, weight decay=1e-4), batch size 32, up to 40 epochs with early stopping (patience 8) on validation macro-F1; the best-validation checkpoint is evaluated once on the untouched test split. **C** denotes sharp-image fusion; **C-Blur** denotes the identical procedure on the blurred-image embedding.

D. Why Frozen Pretrained Encoders

As in prior resource-constrained work, this choice is driven directly by measured hardware constraints (CPU-only, 8GB RAM): batched frozen-DistilBERT extraction measured 46.5–67.6 ms/sample, and batched frozen-ResNet18 extraction measured 63.2–70.5 ms/sample (sharp and blurred conditions, all splits) — tractable only because the encoders are frozen rather than fine-tuned end to end.

VI. EXPERIMENTAL SETUP

Fixed seed (42) throughout. Metrics: accuracy, macro-F1, weighted-F1, and per-class precision/recall/F1 with confusion matrices; macro-F1 is the primary metric given the class imbalance in Section IV. All numbers in Section VII are read directly from results/*.json produced by this project's own pipeline scripts, run end to end on CPU-only, 8GB-RAM hardware, with no numbers invented or adjusted post hoc.

A. Hardware and Software Environment

All experiments ran on a single consumer laptop: CPU-only (no GPU), 8GB RAM, Windows 10. Software: Python 3.13, PyTorch 2.14 (CPU build), torchvision 0.29, Hugging Face Transformers 5.18, and scikit-learn 1.9. This is the same class of constrained hardware used in prior resource-constrained work in this area, reported here in full rather than abstracted away, since it directly motivated the frozen-encoder modelling choices in Section V and the stratified subsampling decision in Section IV.

B. Fusion Training Behaviour

The fusion MLP (C) converged with early stopping at epoch 21 of a maximum 40 (best validation macro-F1 checkpoint retained); the blurred-image fusion model (C-Blur) trained longer before early stopping, at epoch 32 — consistent with the blurred-image embedding carrying a weaker, slower-to-fit signal for the fusion head to exploit, a further behavioural indicator (alongside the headline metrics) that the sharp-image embedding offers the fusion model more immediately usable signal than the blurred one.

VII. RESULTS

Table I reports test-set performance for every model, sharp and blurred image conditions (text-only models are listed once).

TABLE I
RESULTS: TEST-SET PERFORMANCE BY MODEL AND IMAGE CONDITION

Model	Acc.	Macro-F1	Wtd-F1
A1 — TF-IDF + LogReg (text)	42.6%	33.3%	43.7%
A2 — Frozen DistilBERT + LogReg (text)	39.4%	33.4%	42.1%
B1 — CNN from scratch (sharp)	61.1%	25.3%	46.4%
B2 — Frozen ResNet18 + LogReg (sharp)	38.6%	31.0%	42.2%
B2-Blur — Frozen ResNet18 + LogReg (blur)	37.1%	30.7%	39.6%
C — Concat Fusion (text + sharp image)	**52.6%**	**36.6%**	**50.6%**
C-Blur — Concat Fusion (text + blurred image)	41.7%	33.0%	44.4%

RQ1 (does fusion help): yes. Fusion on sharp images (C) clearly outperforms every unimodal model on both headline metrics: +10.0pp accuracy over the best unimodal accuracy (A1) and +3.2pp macro-F1 over the best unimodal macro-F1 (A2). This replicates, on a structurally different data type, the qualitative finding that multimodal fusion improves social-media sentiment classification over either modality alone.

On B1: B1 reaches the highest raw accuracy (61.1%) but the lowest macro-F1 (25.3%) — its confusion matrix shows collapse toward the majority (positive) class, a genuine training-instability failure under this hardware's constrained training budget rather than a real feature-learning advantage; its accuracy alone is not a meaningful comparison point, which is why macro-F1 is treated as primary. Table III makes this concrete: training loss stays pinned near $\ln(3) \approx 1.099$ (i.e., close to uniform-random output) for all 6 epochs, validation macro-F1 is identical at epochs 1, 2, and 6 (0.249, matching pure majority-class prediction), and briefly collapses further at epoch 5 (0.161, accuracy 31.7%) before reverting — the model never escapes a narrow band of degenerate, near-constant predictions.

TABLE III
B1 (CNN FROM SCRATCH) PER-EPOCH TRAINING BEHAVIOUR

Epoch	Train Loss	Val Acc.	Val Macro-F1
1	1.100	0.597	0.249
2	1.099	0.597	0.249
3	1.099	0.586	0.252
4	1.098	0.600	0.256
5	1.102	0.317	0.161
6	1.098	0.597	0.249

A. RQ2: The Blur Ablation

Fig. 1 shows accuracy and macro-F1 for every image-dependent configuration, sharp vs. blurred. Two results stand out. First, B2 is almost unaffected by blurring (accuracy 38.6%→37.1%, −1.4pp; macro-F1 31.0%→30.7%, −0.2pp): the image-only linear classifier was evidently not relying heavily on fine legible-text detail. Second, fusion is substantially affected (accuracy 52.6%→41.7%, −10.9pp; macro-F1 36.6%→33.0%, −3.6pp): most of fusion's advantage over text-only evaporates. C-Blur's accuracy is only 2.3pp above text-only (A2) and its macro-F1 is 0.4pp *below* text-only.

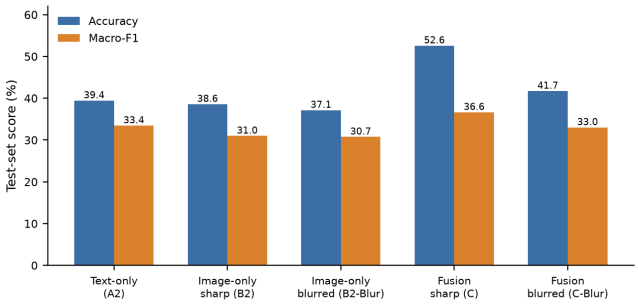


Fig. 1. Sharp vs. blurred accuracy and macro-F1, by model.

Taken together, this is the paper's central finding: the multimodal fusion gain measured here is substantially mediated by the image channel's access to legible embedded meme text, not by independent visual sentiment cues that a linear probe on frozen ResNet18 features can already detect. The fusion MLP evidently extracts residual text-correlated signal from the sharp-image embedding that the simpler B2 probe does not surface on its own — and that signal specifically, not general visual composition, is what the blur ablation removes.

B. Confusion Matrix: Fusion, Sharp vs. Blurred

Table II shows the test-set confusion matrix for C (sharp) and C-Blur. The clearest shift under blurring is the positive class: C correctly recovers 159 of 214 positive memes (74.3% recall) but C-Blur recovers only 105 (49.1% recall), with many positive memes instead misclassified as neutral (72 under blur vs. 26 sharp) — i.e., once embedded text is illegible, the fusion model loses much of its ability to confidently identify positive memes specifically, which is consistent with positive-sentiment memes in this dataset leaning heavily on readable caption wording (e.g., explicit celebratory or congratulatory text) that a blurred image can no longer supply.

TABLE II
TEST CONFUSION MATRICES: C (SHARP) VS. C-BLUR

(a) C — fusion, sharp image

True \ Pred	Neg.	Neu.	Pos.
Neg.	8	4	15
Neu.	21	17	71
Pos.	29	26	159

(b) C-Blur — fusion, blurred image

True \ Pred	Neg.	Neu.	Pos.
Neg.	6	10	11
Neu.	24	35	50
Pos.	37	72	105

C. RQ3: Modality Conflict Analysis

Text-only (A2) and image-only (B2, sharp) predictions disagree on 63.1% of test memes (221/350) — higher than the 53.3% disagreement rate reported for photographic tweet images, consistent with meme images and their embedded captions being a less independent signal pair. On this disagreement subset, fusion outperforms both unimodal models (51.6% vs. 37.1% text-only, 35.7% image-only). Unlike the photographic-image pattern — a small regression where modalities agree — fusion here also outperforms both unimodal models on the agreement subset (54.3% vs. 43.4% for both). We attribute this to the same cause as the blur-ablation finding: because meme "image" information is partly text-derived, the fusion head gains a more consistently useful signal than in the photographic-image case.

Per-class: fusion (C) substantially improves the majority positive class over text-only (F1 69.3% vs. 49.6%) and the minority negative class (F1 18.8% vs. 17.0%), but regresses on neutral (F1 21.8% vs. 33.8%, recall 35.8%→15.6%) — the same majority-reinforcing, minority-adjacent-class cost pattern reported for fusion on photographic images, replicated here on a different dataset and task framing.

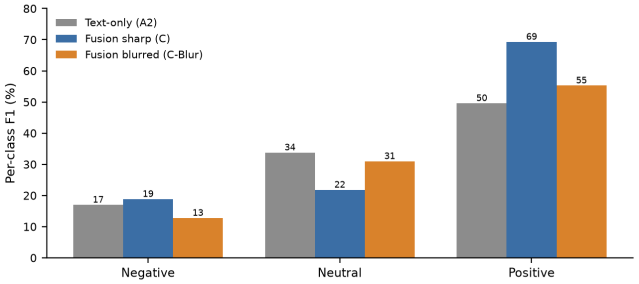


Fig. 2. Per-class F1: text-only vs. fusion sharp vs. fusion blurred.

VIII. DISCUSSION, ETHICS, AND LIMITATIONS

A. Practical Implication

Any system fusing "image" and "text" modalities for meme-like content (memes, captioned infographics, text screenshots) should not assume the image channel is an independent evidence source. Reported multimodal gains on such content may substantially reflect the image model re-deriving the same text signal rather than genuine cross-modal understanding; evaluations that care about this distinction should include a leakage ablation of this kind.

B. Non-Clinical, Non-Moderation Scope

This study measures sentiment-label prediction accuracy only; it makes no claims about detecting humor, sarcasm, offensiveness, or hateful content, despite the dataset providing such labels, and should not be read as a validated content-moderation tool.

C. Threats to Validity

Internal validity: all reported numbers come from a single run per model at a fixed seed (42); we do not report variance across multiple seeds, so small differences between models (e.g., B2 vs. B2-Blur's 1.4pp accuracy gap) should be read with appropriate caution, while the large differences central to this paper's claims (C vs. C-Blur's 10.9pp accuracy gap) are unlikely to be seed noise given their size relative to the small-difference case observed elsewhere in the same table. **Construct validity:** the blur ablation is a proxy for "text illegibility," not a direct text-region mask; Section VIII-D already discusses this as a limitation rather than a resolved concern. **External validity:** findings are specific to this dataset, label scheme, and model family, and should not be assumed to transfer to other meme corpora, languages, or encoder architectures without further testing.

D. Key Limitations

(1) the 3,500-sample stratified subsample is smaller than the full 6,992-meme dataset; results may shift at full scale. (2) the 5-to-3-class sentiment collapse discards intensity information. (3) the Gaussian blur removes all fine high-frequency detail, not exclusively text; some measured effect could reflect loss of other fine visual detail, though the meme format's text-dominant high-frequency content makes text the most plausible primary contributor. (4) the dataset is English-language, non-codemixed memes from a fixed collection period and may not generalize to other languages, formats, or eras. (5) B1's training instability limits how much weight its result should carry. (6) CPU-only compute constrained the from-scratch CNN's training budget and resolution, documented rather than hidden.

IX. CONCLUSION AND FUTURE WORK

Within a real, resource-constrained CPU-only environment, this study finds multimodal fusion improves meme sentiment classification over unimodal baselines (RQ1), replicating the pattern reported for photographic social-media images. However, a direct blur ablation shows a substantial share of this gain is attributable to the image channel's access to legible embedded text rather than independent visual cues (RQ2) — the gain collapses to near text-only performance once that text is illegible. The modality-conflict pattern also differs from prior photographic-image findings: fusion helps on both the disagreement and agreement subsets here (RQ3). Future work: a direct OCR-region-masking leakage estimate rather than global blurring; evaluating whether this pattern holds for the dataset's humor/sarcasm/offensiveness labels; and testing whether an image encoder explicitly trained to ignore text regions recovers a meaningful independent-visual-signal multimodal gain.

CODE AND DATA AVAILABILITY

All pipeline scripts (data export, feature extraction, model training, analysis, and figure/PDF generation), the fixed random seed, and the raw results/*.json outputs underlying every number and figure in this paper are retained alongside this report for independent re-execution. The Memotion dataset is publicly available through the SemEval-2020 Task 8 organizers and through third-party mirrors, including the one used here; we did not redistribute any dataset files ourselves.

ACKNOWLEDGMENT

Dataset: Sharma et al. (2020, SemEval-2020 Task 8 / Memotion) [3], accessed via the Ahren09/MMSoc_Memotion Hugging Face mirror. Pretrained models: DistilBERT and ResNet18, accessed via Hugging Face Transformers and torchvision respectively.

APPENDIX: HYPERPARAMETER SUMMARY

Model	Key Hyperparameters
A1 TF-IDF+LogReg	1-2 grams, max 10k features, sublinear TF; LogReg: C=1.0, class_weight=balanced, max_iter=2000
A2 DistilBERT+LogReg	distilbert-base-uncased, mean-pooled, max_len=96; LogReg: C=1.0, class_weight=balanced
B1 CNN (scratch)	4 conv layers (16/32/64/64 ch.), 96x96 input, Adam lr=1e-3, batch=32, 6 epochs, dropout=0.3, class-weighted CE
B2 ResNet18+LogReg	ImageNet-pretrained, frozen, 224x224 input, 512-dim penultimate layer; LogReg: C=1.0, class_weight=balanced
B2-Blur (ablation)	Identical to B2, but Gaussian blur radius=14px applied pre-resize
C / C-Blur Fusion MLP	Input 1280-dim (768+512), hidden 256{→}128{→}3, dropout 0.4/0.3, Adam lr=1e-3, weight_decay=1e-4, batch=32, max 40 epochs, early stop patience=8 on val macro-F1

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